考虑接地的导体边界，放入一点电荷，产生电势 $\Gamma(x-y)$ ← fundamental sol.

导体表面感应电荷重新分布，产生 $-h(x,y)$，而 $G(x,y)=\Gamma(x-y)-h(x,y)$ $\Rightarrow \begin{cases} -\Delta_x G(x,y)=\delta_y \\ G(x,y)=0 \end{cases}$

Step 1：解决点源问题：y为点源

$$\begin{cases} \Delta_x G(x,y)=\delta_y(x) & x \text{ in } \Omega \\ G(x,y)=0 & x \text{ in } \partial\Omega \end{cases}$$

$\Gamma(x-y)-h(x,y)$

Step 2：解决边界零值问题：

$$\begin{cases} -\Delta_x u=f(x) & \text{in } \Omega \\ u=0 & \text{in } \partial\Omega \end{cases}$$

then $u(x)=\int_\Omega f(y)\,G(x,y)\,dy$

Step 3：解决边界贡献：

$$\begin{cases} -\Delta_x u=f(x) & \text{in } \Omega \\ u=g & \text{in } \partial\Omega \end{cases}$$

then $u(x)=\int_\Omega f(y)\,G(x,y)\,dy+\int_{\partial\Omega} -\partial_{n_y} G(x,y)\,g(y)\,dS_y$

Case 2) : Ω is bounded region in $\mathbb{R}^n$

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{in } \partial\Omega \end{cases}$$

We can start with

$$\begin{cases} -\Delta u = f & x \in \Omega \\ u = 0 & x \in \partial\Omega \end{cases}$$

(Hope : Similar to $\Omega = \mathbb{R}^n$, we want to find $G(x,y)$. $\int_\Omega G(x,y) f(y)\,dy$ solve the eq.)

Such $G(x,y)$ is called Green - Function

Informally, take $f(x) = \delta_0(x-y)$, then

$$\begin{cases} -\Delta_x G(x,y) = \delta_0(x-y) & x \in \Omega \\ G(x,y) = 0 & x \in \partial\Omega \end{cases}$$

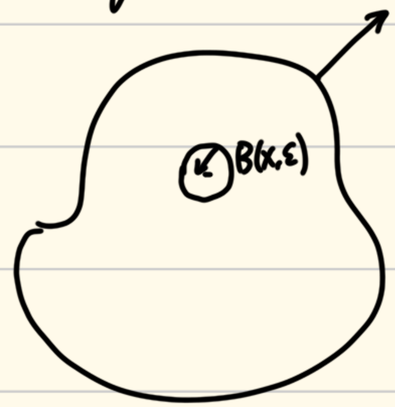
$\Gamma(x-y)$ solves 1 but fails at 2, Guess $G(x,y) = \Gamma(x-y) + \phi(x,y)$ → as a corrector

corrector ϕ satisfy :

$$\begin{cases} \Delta_x \phi(x,y) = 0 & x \in \Omega \\ \phi(x,y) = \Gamma(x-y) & x \in \partial\Omega \end{cases}$$

Suppose we find such $\phi(x,y)$. the we get a formula for Poisson eq.

Now we try to give rigorous derivation of Green's function and Poisson's Eq.



$$\Omega_\varepsilon = \Omega \setminus B(x,\varepsilon)$$

$$\int_{\Omega_\varepsilon} u(y)\,\Delta_y \Gamma(y-x) - \Delta_y u(y)\,\Gamma(y-x)\,dy$$

$$= \int_{\Omega_\varepsilon} u(y)\,\frac{\partial \Gamma}{\partial n}(y-x) - \frac{\partial u}{\partial n}(y)\cdot \Gamma(y-x)\,dS(y)$$

$$= \left(\int_{\partial\Omega} + \int_{\partial B(x,\varepsilon)}\right) u(y)\,\frac{\partial \Gamma}{\partial n}(y-x) - \frac{\partial u}{\partial n}(y)\cdot \Gamma(y-x)\,dS(y)$$

$$\left|\int_{\partial B(x,\varepsilon)} \frac{\partial u}{\partial n}\,\Gamma\,dS(y)\right| \begin{cases} \le C\int_{\partial B(x,\varepsilon)} \log\varepsilon\,dS(y) \le \widetilde{C}\,\varepsilon\log\varepsilon \\ \le C\int_{\partial B(x,\varepsilon)} \varepsilon^{2-n}\,dS(y) \le \overline{C}\,\varepsilon \end{cases} \to 0$$

$$n = -\frac{y-x}{|y-x|}, \quad \frac{\partial \Gamma}{\partial \vec{n}} = \frac{\partial \Gamma}{\partial y}\cdot\vec{n} = \begin{cases} -\frac{1}{2\pi}\cdot\frac{1}{|y-x|} & n=2 \\ \frac{1}{n\alpha(n)}\cdot\frac{1}{|y-x|^{n-1}} & n\ge 3 \end{cases} = \frac{1}{n\alpha(n)}\cdot\frac{1}{|y-x|^{n-1}}$$

$$\int_{\partial B(x,\varepsilon)} u\cdot\frac{1}{n\alpha(n)}\cdot\frac{1}{\varepsilon^{n-1}}\,dS(y) = \fint u\,dS(y) \to u(x)$$

How ?

Hence : let $\varepsilon\to 0$ $\quad u(x) = \int_\Omega f(y)\,\Gamma(y-x)\,dy - \int_{\partial\Omega} g(y)\,\frac{\partial \Gamma}{\partial n}\,dS(y) + \int_{\partial\Omega} \frac{\partial u}{\partial n}\,\Gamma\,dS(y)$ (Eq. 1)

To kill the last term we introduce the corrector $\phi^x(y)$:

$$\begin{cases} -\Delta_y \phi^x(y) = 0 & y \in \Omega \\ \phi^x(y) = \Gamma(x-y) & y \in \partial\Omega \end{cases}$$

Apply Green's 2nd Identity : $\int_\Omega \underbrace{u\Delta\phi}_{=0} - \phi\Delta u = \int_{\partial\Omega} u\frac{\partial\phi}{\partial n} - \underbrace{\phi}_{=\Gamma}\frac{\partial u}{\partial n}\, dS(y)$

$$\Rightarrow 0 = \int_{\partial\Omega} u\frac{\partial\phi}{\partial n} - \Gamma\frac{\partial u}{\partial n}\, dS(y) + \int_\Omega \phi\Delta u\, dy \quad (\text{Eq. } 2)$$

$$(\text{Eq. } 1) + (\text{Eq. } 2) \Rightarrow u(x) = \int_\Omega f(y)\Gamma(y-x)\,dy - \int_{\partial\Omega} g(y)\frac{\partial\Gamma}{\partial n}\,dS(y) + \int_{\partial\Omega} u\frac{\partial\phi}{\partial n}\,dS(y) - \int_\Omega \phi f\, dy$$

$$= \int_\Omega f(y)\left(\Gamma(y-x) - \phi^x(y)\right)dy - \int_{\partial\Omega} g\frac{\partial}{\partial n}\left(\Gamma(x-y) - \phi^x(y)\right)dS(y)$$

Summarize :

Def : $\begin{cases} -\Delta_y \phi^x(y) = 0 & y \in \Omega \\ \phi^x(y) = \Gamma(x-y) & y \in \partial\Omega \end{cases}$ we call $\phi^x(y)$ the corrector

Def : $G(x,y) = \Gamma(x-y) - \phi^x(y)$ the Green function on domain Ω

$G(x,y) : \bar{\Omega} \times \bar{\Omega} \setminus \{x=y\} \to \mathbb{R}$

$$\begin{cases} -\Delta_y G(x,y) = \delta(x-y) & y \in \Omega \\ G(x,y) = 0 & y \in \partial\Omega \end{cases}$$

Fact : $u(x) = \int_\Omega f(y)\, G(x,y)\, dy - \int_{\partial\Omega} g(y) \frac{\partial G}{\partial n} dy$ solves $\begin{cases} -\Delta u = f & x \in \Omega \\ u = g & x \in \partial\Omega \end{cases}$

Lemma: $\forall a,b\in\Omega,\ a\neq b,\ G(a,b)=G(b,a)$



Pf: Consider $f(x)=G(a,x),\quad g(x)=G(b,x)$

$$0=\int_{\Omega_\varepsilon}\Delta f\,g-f\,\Delta g\,dx$$

$$=\underbrace{\int_{\partial\Omega}\frac{\partial f}{\partial n}g-\frac{\partial g}{\partial n}f\,dS(x)}_{=0}+\int_{\partial B(a,\varepsilon)}\frac{\partial f}{\partial n}g-\frac{\partial g}{\partial n}f\,dS(x)-\int_{\partial B(b,\varepsilon)}\frac{\partial f}{\partial n}g-\frac{\partial g}{\partial n}f\,dS(x)$$

$$\int_{\partial B(a,\varepsilon)}\frac{\partial f}{\partial n}g\,dS(x)=\int_{\partial B(a,\varepsilon)}\underbrace{\frac{1}{n\alpha(n)\,\varepsilon^{n-1}}}_{\to g(a)}g\,dS(x)-\int_{\partial B(a,\varepsilon)}\underbrace{\frac{\partial\phi^x(y)}{\partial n}}_{\to 0}g(x)\,dS(x)$$

$$\left|\int_{\partial B(a,\varepsilon)}f\frac{\partial g}{\partial n}\,dS(x)\right|=\begin{cases}n=2: & \left|\int_{\partial B(a,\varepsilon)}\left(-\frac{1}{2\pi}\log\varepsilon-\phi^a(x)\right)\frac{\partial g}{\partial n}\,dS(x)\right|\leq C(\varepsilon\log\varepsilon+\varepsilon)\to 0\\ n\geq 3: & \left|\int_{\partial B(a,\varepsilon)}\left(\frac{1}{n\alpha(n)(2-n)}\,\frac{1}{\varepsilon^{n-2}}-\phi^a(x)\right)\frac{\partial g}{\partial n}\,dS(x)\right|\leq C(\varepsilon+\varepsilon^{n-1})\to 0\end{cases}$$

($\int_{\partial B(a,\varepsilon)}\frac{\partial f}{\partial n}g\,dS(x)\to 0$)

Hence, let $\varepsilon\to 0$: $g(a)=f(b)$ $\quad G(a,b)=G(b,a)$ #

Ex 1 : $\Omega = \mathbb{R}^n_+ = \{(x_1, \cdots x_n) \in \mathbb{R}^n,\ x_n > 0\}$ $\quad \partial\Omega = \{(x_1 \cdots x_{n-1}, 0)\} \cong \mathbb{R}^{n-1}$

We seek for $\phi^x(y)$ $\quad \begin{cases} -\Delta_y \phi^x(y) = 0 & y \in \mathbb{R}^n_+ \\ \phi^x(y) = \Gamma(x-y) & y \in \partial\mathbb{R}^n_+ \end{cases}$

$\forall x \in \mathbb{R}^n_+$, define $\hat{x} = (x_1 \cdots x_{n-1}, -x_n)$ be the reflection of x

$\Gamma(y-\hat{x}) = \Gamma(y_1 - x_1, y_2 - x_2, \cdots y_n + x_n)$ $\quad -\Delta_y \Gamma(y-\hat{x}) = 0$

$\Gamma(y-\hat{x})\big|_{y \in \partial\mathbb{R}^n_+} = \Gamma(x-y)$

hence we can take $\phi^x(y) = \Gamma(y-\hat{x})$, $G(x,y) = \Gamma(y-x) - \Gamma(y-\hat{x})$

We consider a special case $\quad \begin{cases} -\Delta u = 0 & x \in \mathbb{R}^n_+ \\ u = g & x \in \partial\mathbb{R}^n_+ \end{cases}$

then $u(x) = \int_{\partial\Omega} g(y) \frac{\partial G(x,y)}{\partial n} dS(y)$

$$\frac{\partial G}{\partial n} = -\frac{\partial G}{\partial y_n} = -\partial_{y_n} \Gamma(y-x) + \partial_{y_n} \Gamma(y-\tilde{x}) = \frac{(-1)}{n\alpha(n)} \frac{y_n - x_n}{|y-x|^n} - \frac{(-1)}{n\alpha(n)} \frac{y_n - \tilde{x}_n}{|y-\tilde{x}|^n}$$

$$\left.\frac{\partial G}{\partial y_n}\right|_{y \in \partial \mathbb{R}^n_+} = -\frac{1}{n\alpha(n)} \cdot \frac{x_n - \tilde{x}_n}{|y-x|^n} = \frac{2x_n}{n\alpha(n)\,|y-x|^n}$$

$$\Rightarrow u(x) = \int_{\partial \mathbb{R}^n_+} \frac{2x_n}{n\alpha(n)} \frac{g(y)}{|y-x|^n} \, dS(y) = \int_{\mathbb{R}^{n-1}} \frac{2x_n}{n\alpha(n)} \frac{g(y)}{\left[(y_1-x_1)^2 + \cdots + x_n^2\right]^{\frac{n}{2}}} \, dy$$

$$K(x,y) = \frac{2x_n}{n\alpha(n)\,|y-x|^n}, \quad u(x) = \int_{\mathbb{R}^{n-1}} K(x,y)\, g(y)\, dy$$

Thm: $g \in C_b(\mathbb{R}^{n-1})$ then $u = \int_{\partial \mathbb{R}^n_+} K(x,y)\, g(y)\, dS(y)$ satisfies

① $u \in C^\infty(\mathbb{R}^n_+)$ & bounded ② $-\Delta u = 0$

③ $\forall x_0 \in \partial \mathbb{R}^n_+$, $x \in \mathbb{R}^n_+$, $\lim_{x \to x_0} u(x) = g(x_0)$

Pf: ① By direct computation, $\int_{\partial \mathbb{R}^n_+} K(x,y)\, dS(y) = 1$, hence u is bounded

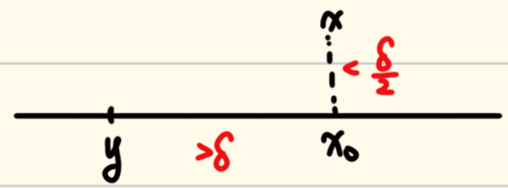
Also $K(x,y)$ is smooth w.r.t. x, hence $u \in C^\infty(\mathbb{R}^n_+)$

② $-\Delta_x K(x,y) = 0 \Rightarrow -\Delta u = 0$

③ Fix $x_0 \in \partial \mathbb{R}^n_+$, $\forall \varepsilon$, $\exists \delta_1 > 0$ s.t. $\forall |y - x_0| < \delta_1$, $y \in \partial \mathbb{R}^n_+$, $|g(y) - g(x)| < \frac{\varepsilon}{2}$

$$u(x) - g(x_0) = \int_{\partial \mathbb{R}^n_+} K(x,y)\,(g(y) - g(x_0)) = \int_{|y-x_0|<\delta} + \int_{|y-x_0|>\delta}$$

$$\left|\int_{|y-x_0|<\delta}\right| \le \frac{\varepsilon}{2} \int_{\partial \mathbb{R}^n_+} K(x,y) = \frac{\varepsilon}{2} \qquad \left|\int_{|y-x_0|>\delta}\right| \le C \int_{|y-x_0|>\delta} K(x,y)\, dy$$



若 $|x - x_0| < \frac{\delta_1}{2}$, 则 $|y - x| \ge \frac{1}{2}|y - x_0| \Rightarrow \left|\int_{|y-x_0|>\delta}\right| \le C \int_{|y-x_0| \ge \delta_1} \frac{x_n}{\left(\frac{1}{2}|y-x_0|\right)^n}\, dy \le \tilde{C} \int_{r \ge \delta_1} \frac{x_n}{r^n} \cdot r^{n-2}\, dr = \tilde{C}\, x_n \frac{1}{\delta_1} \le \frac{\varepsilon}{2}$

取 $x_n \le \frac{\varepsilon \delta_1}{2\tilde{C}}$

取 $|x - x_0| \le \delta = \min\left\{\frac{\delta_1}{2}, \frac{\varepsilon \delta_1}{2\tilde{C}}\right\}$, $\left|\int_{|y-x_0|>\delta}\right| \le \frac{\varepsilon}{2}$ $\Rightarrow u(x) \to g(x_0)$

Section 2.2 Laplace Eq.

Ex2 $\Omega = B(0,R)$ $\begin{cases} -\Delta u = f & x \in \Omega \\ u|_{\partial\Omega} = g \end{cases}$ First consider $R=1$

Search for corrector : $\begin{cases} -\Delta_y \phi^x(y) = 0 & \text{in } B(0,1) \\ \phi^x(y) = \Gamma(x-y) & \text{in } \partial B(0,1) \end{cases}$

Consider $x^* = \frac{x}{|x|^2}$, $x^* \cdot x = 1$

Take $\phi^x(y) = \Gamma\left(|x|(y-x^*)\right)$. $G(x,y) = \Gamma(y-x) - \Gamma\left(|x|(y-x^*)\right)$

So solution $u(x) = -\int_{\partial B_1} g(y) \frac{\partial G(x,y)}{\partial n_y} dS_y$

Compute : $\frac{\partial G}{\partial n_y} = \frac{\partial G}{\partial y} \cdot \frac{y}{|y|} = \frac{1}{n\alpha(n)} \left(\frac{x-y}{|y-x|^n} - \frac{x^*-y}{|x^*-y|^n} \cdot \frac{|x|^2}{|x|^n} \right) \cdot \frac{y}{|y|}$ $(n \geq 3)$

$\Rightarrow \left.\frac{\partial G}{\partial n_y}\right|_{|y|=1} = \frac{1}{n\alpha(n)} \frac{(|x|^2-1)}{|y-x|^n}$

$\Rightarrow u(x) = -\int_{\partial B_1} \frac{1}{n\alpha(n)} \frac{1-|x|^2}{|y-x|^n} g(y)\, dS(y)$

Back to $B(0,R)$: Thm: $g: \partial B(0,R) \to \mathbb{R}$ continuous function, $u = \int_{\partial B(0,R)} K_R(x,y)\, g(y)\, dS(y)$ satisfy

① u bounded & C^∞ ② $-\Delta u = 0$ in $B(0,R)$

Here: $K_R(x,y) = \dfrac{R^2 - |x|^2}{n\alpha(n) R |y'-x'|^2}$

2.2.4 Estimates

a) L^∞ - estimate

Thm: $\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{in } \partial\Omega \end{cases}$, u is a $C^2(\Omega) \cap C(\bar{\Omega})$, then we have

$\sup_{x \in \bar{\Omega}} |u(x)| \le \sup_{\partial\Omega} |g| + C \sup_{\Omega} |f|$ C is a constant of dim & diam(Ω)

Pf: Easy case. $f=0$: Apply maximal principle.

Let's find a function $v(x)$, $-\Delta v = 1$

Put Ω in a large ball $B_d(x_0)$, consider $v(x) = \frac{1}{2n}(d^2 - |x-x_0|^2) \le \frac{d^2}{2n}$

Denote $F = \sup|f|$, $G = \sup|g|$ and consider $w_1(x) = u(x) - G - Fv(x)$

Then $\begin{cases} -\Delta w_1(x) = -\Delta u + F\Delta v(x) = f - F \le 0 \\ w_1(x)|_{\partial\Omega} = g - G - Fv(x)|_{\partial\Omega} = g - G \le 0 \end{cases}$

Since subharmonic func. reaches max on bdry.

$\Rightarrow w_1(x) \le 0 \Rightarrow u(x) \le G + Fv(x) \le G + F \cdot \frac{d^2}{2n}$

Similarly, consider $w_2(x) = -u(x) - G - Fv(x) \Rightarrow w_2(x) \le 0 \Rightarrow u(x) \ge -G - F \cdot \frac{d^2}{2n}$

Hence $\sup|u(x)| \le G + F \cdot \frac{d^2}{2n}$

Corollary : $\begin{cases} -\Delta u_i = f_i \\ u_i = g_i \end{cases}$ then $\sup |u_i - u_j| \le \sup |g_i - g_j| + C \sup |f_i - f_j|$

b) Energy Estimate

Thm : $\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{in } \partial\Omega \end{cases}$ $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ is a sol.

Then : $\int_\Omega |Du|^2 dx \le C \int_\Omega |f|^2 dx$, $C = C(\dim, \operatorname{diam}(\Omega))$

Pf : $\int_\Omega -\Delta u \cdot u = \int_\Omega fu$, $\nabla \cdot (u \nabla u) = |\nabla u|^2 + u \Delta u$

$\Rightarrow \int_\Omega |Du|^2 dx - \int_\Omega \nabla \cdot (u \nabla u) dx = \int_\Omega fu$

$\Rightarrow \int_\Omega |Du|^2 dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} u \, dS = \int_\Omega fu$

$\Rightarrow \int_\Omega |Du|^2 dx = \int_\Omega fu \, dx \le \frac{\varepsilon}{2} \int_\Omega u^2 dx + \frac{1}{2\varepsilon} \int_\Omega f^2 dx$

Lemma (Poincare Ineq.) Ω bounded

① $u \in C_0^1(\Omega) = \{ u \in C^1(\Omega),\ \text{supp } u \text{ compact and } \subset \Omega \}$

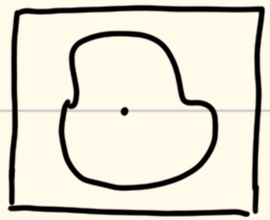
then $\int_\Omega |u|^2 dx \leq C \int_\Omega |Du|^2 dx$

② $u \in C^1(\bar{\Omega})$ & $u = 0$ in $\partial\Omega$

then $\int_\Omega |u|^2 dx \leq C \int_\Omega |Du|^2 dx$

Pf of Lemma :

① extend $\tilde{u} = \begin{cases} u & x \in \bar{\Omega} \\ 0 & x \notin \Omega \end{cases}$ $\tilde{u} \in C^1(\mathbb{R}^n)$



put Ω in a large box D, WLOG $D = [-d, d]^n$ by assumming $0 \in \Omega$

$\forall\ x = (x_1, \cdots x_n) \in \Omega$

$$\tilde{u}(x_1, \cdots x_n) = \tilde{u}(x_1, \cdots x_n) - \tilde{u}(x_1, \cdots x_{n-1}, -d) = \int_{-d}^{x_n} \partial_{x_n} \tilde{u}(x_1 \cdots x_{n-1}, \xi)\, d\xi$$

$$\Rightarrow |\tilde{u}(x_1, \cdots x_n)|^2 \overset{C\text{-}S}{\leq} \int_{-d}^{x_n} 1\, d\xi \cdot \int_{-d}^{x_1} (\partial_{x_n} \tilde{u})^2 (x_1 \cdots x_{n-1}, \xi)\, d\xi \leq 2d \int_{-d}^{d} (\partial_{x_n} \tilde{u})^2 (x_1 \cdots x_{n-1}, \xi)\, d\xi$$

Hence by Fubini $$\int_{-d}^{d}\cdots\int_{-d}^{d}|\tilde{u}(x)|^2dx=\int_{-d}^{d}\cdots\int_{-d}^{d}\left(\int_{-d}^{d}|\tilde{u}(x_1,\cdots x_{n-1},x_n)|^2dx_n\right)dx'$$

$$\leq 2d\int_{-d}^{d}\cdots\int_{-d}^{d}\left(\int_{-d}^{d}|\partial_{x_n}\tilde{u}(x_1,\cdots x_{n-1},\xi)|^2d\xi\right)dx_n\,dx'$$

$$=(2d)^2\int_{-d}^{d}\cdots\int_{-d}^{d}|\partial_{x_n}\tilde{u}(x_1,\cdots x_{n-1},\xi)|^2d\xi\,dx'$$

$$\leq(2d)^2\int_{\Omega}|Du|^2dx$$



② $\forall\,(x_1\cdots x_{n-1})=x'\in\mathbb{R}^{n-1}$

$I_{x'}=\{x_n\in\mathbb{R}\mid(x_1\cdots x_{n-1},x_n)\in\Omega\}\subseteq\mathbb{R}$ and $I_{x'}$ is open!

$I_{x'}=\bigcup_{j=1}^{\infty}(a_j,b_j)$

since $u(x',t)-u(x',a)=\int_{a_j}^{t}\partial_{x_n}(x',\xi)d\xi\overset{C.S.}{\Rightarrow}|u(x',t)|^2\leq d\int_{a_j}^{t}(\partial_{x_n}u)^2(x',\xi)\,d\xi$

$$\Rightarrow\int_{a_j}^{b_j}|u(x',t)|^2dt\leq d\int_{a_j}^{b_j}\int_{a_j}^{t}|\partial_{x_n}u(x',\xi)|^2d\xi\,dt\leq d^2\int_{a_j}^{b_j}|\partial_{x_n}u(x',\xi)|^2d\xi$$

$$\Rightarrow\int_{I_{x'}}|u(x',t)|^2dt\leq d^2\int_{I_{x'}}|\partial_{x_n}u(x',t)|^2dt$$

(Fubini) $$\Rightarrow\int_{\Omega}|u|^2dx\leq d^2\int_{\Omega}|Du|^2d\lambda$$

Back to thm: $\int_\Omega |Du|^2 dx = \int_\Omega fu\, dx \leq \frac{\varepsilon}{2}\int_\Omega u^2 dx + \frac{1}{2\varepsilon}\int_\Omega f^2 dx$

$\leq \frac{\varepsilon C}{2}\int_\Omega |Du|^2 dx + \frac{1}{2\varepsilon}\int_\Omega f^2 dx$

Pick $\varepsilon = \frac{1}{C} \Rightarrow \int_\Omega |Du|^2 dx \leq C\int_\Omega f^2 dx$

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